



MRC Cognition
and Brain
Sciences Unit



UNIVERSITY OF
CAMBRIDGE

Functional MRI

pre-processing and analysis

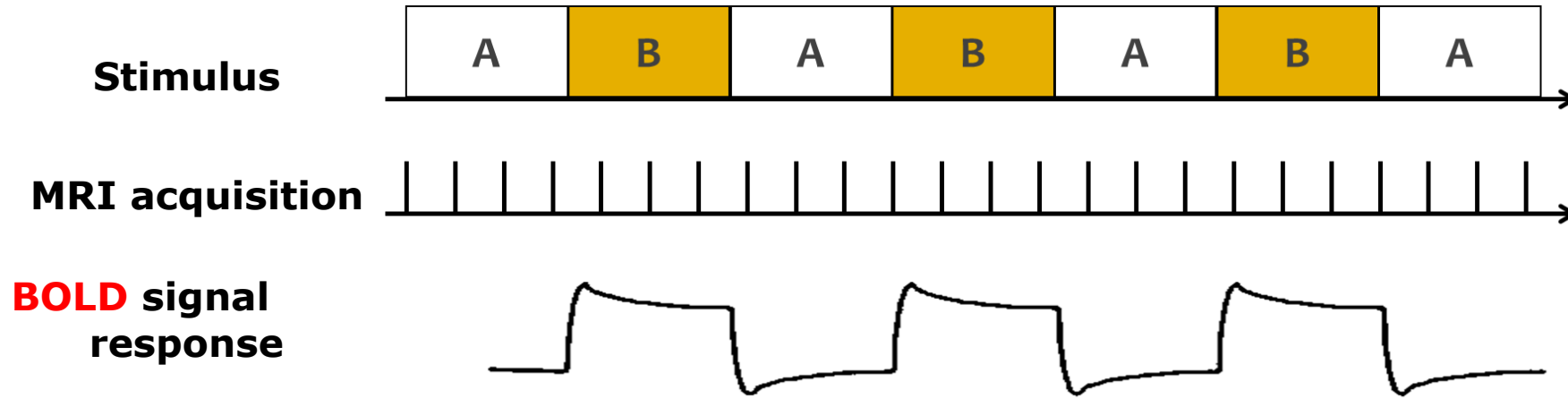
Dace Apšvalka
[Datza]

Functional MRI (fMRI)

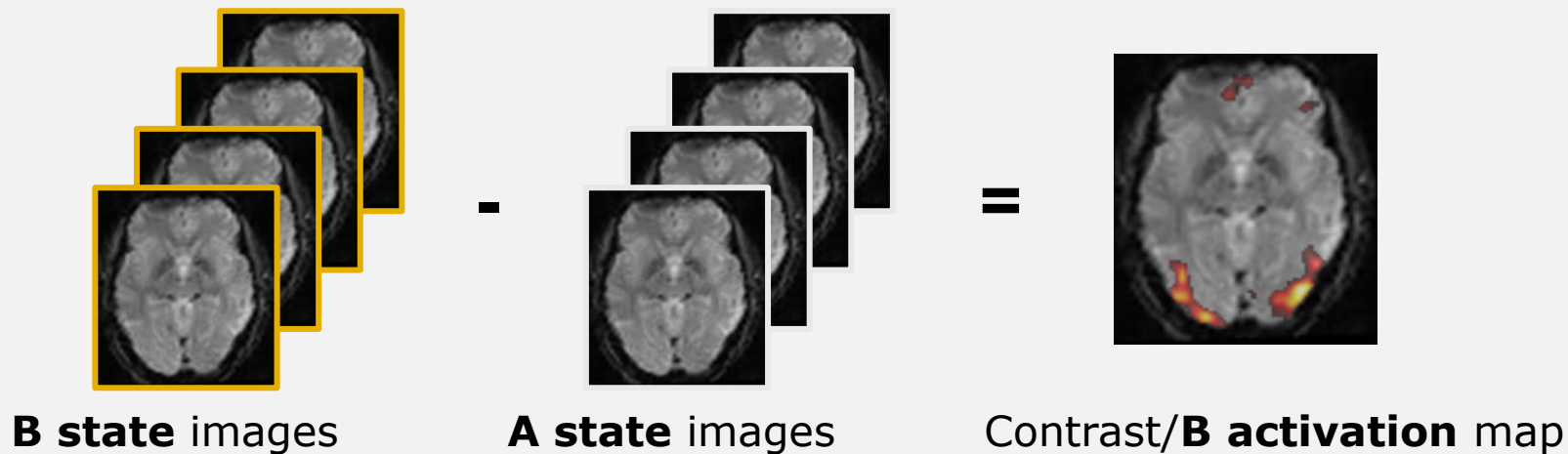
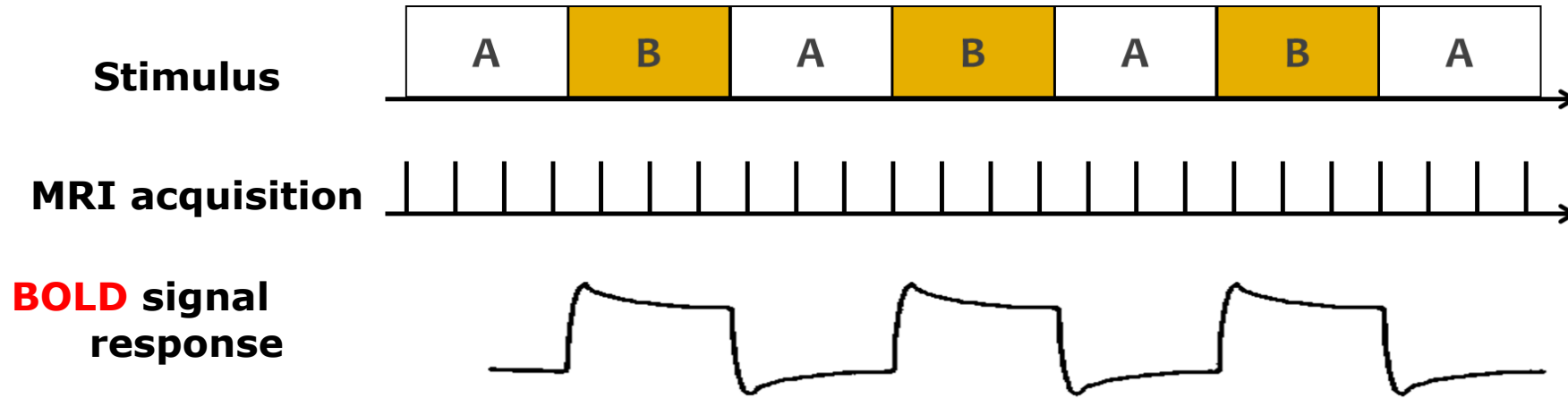
Stimulus



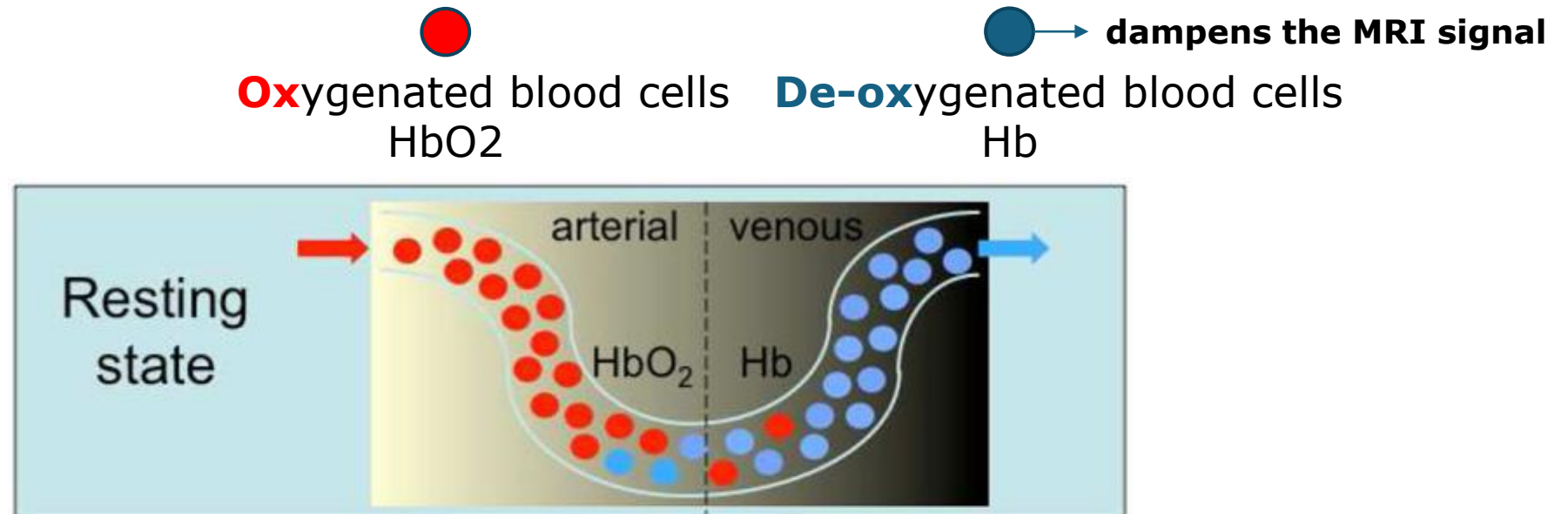
Functional MRI (fMRI)



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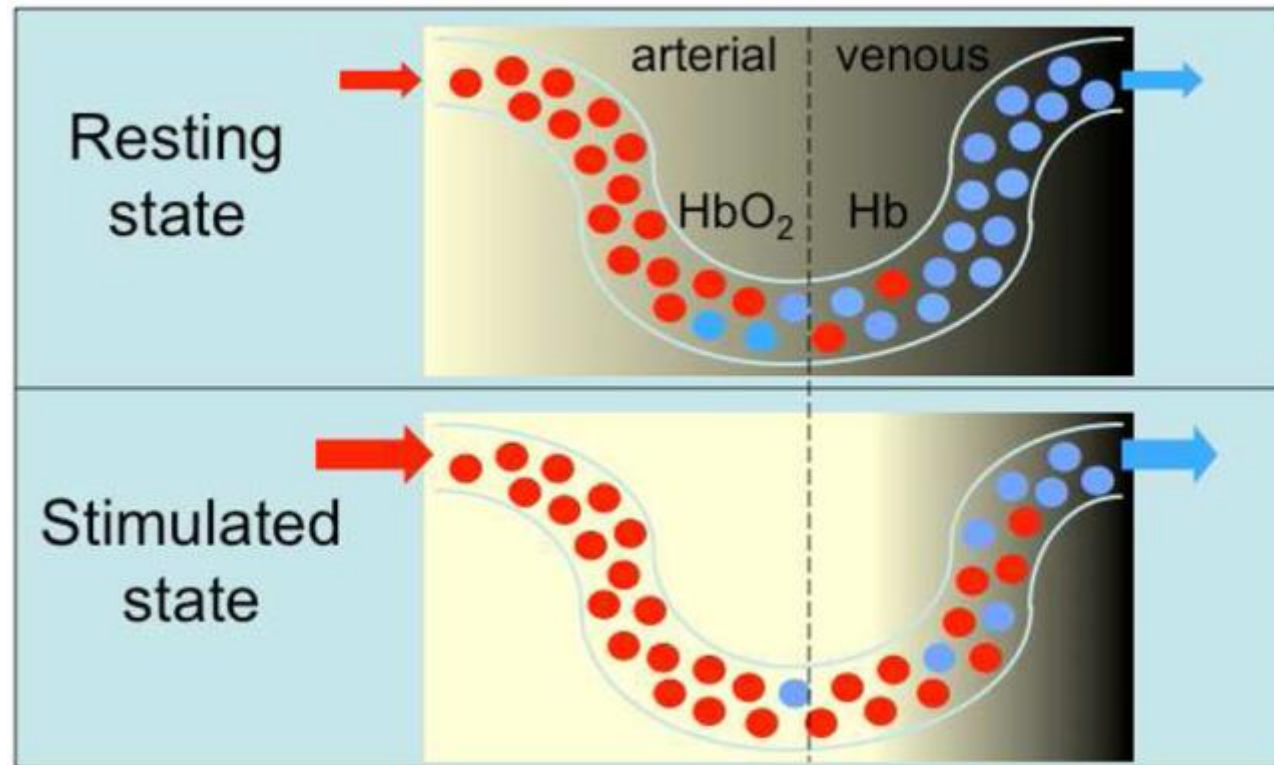


Blood Oxygen Level-Dependent (BOLD) signal



Blood Oxygen Level-Dependent (BOLD) signal

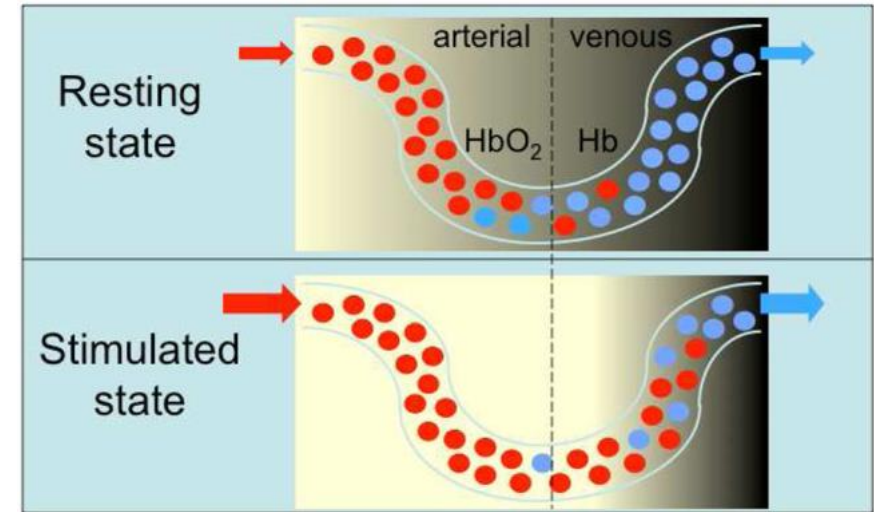
 **O**xxygenated blood cells HbO_2  **De-o**xxygenated blood cells Hb  dampens the MRI signal



Neural activity-induced increase in blood flow **sweeps the “de-ox” away**, causing an MRI signal increase

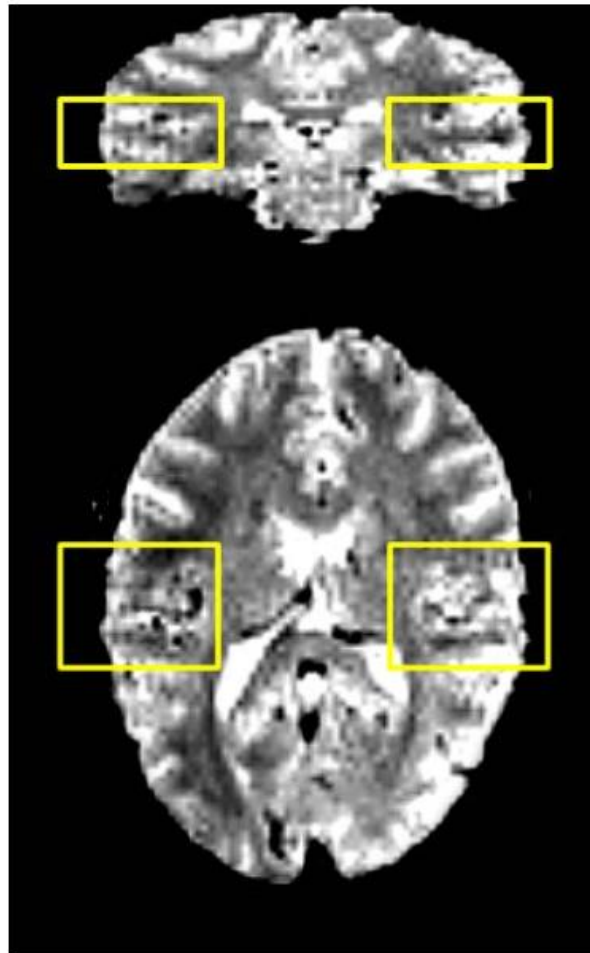
Blood Oxygen Level-Dependent (BOLD) signal

- This difference in the magnetic properties of **de-oxygenated** and **oxygenated** Hb is used in BOLD fMRI to create BOLD contrast in images – reflecting activity in different brain regions
- By controlling for all other factors, any observed **differences in the BOLD signal** are inferred to be due to **differences in neuronal activity**

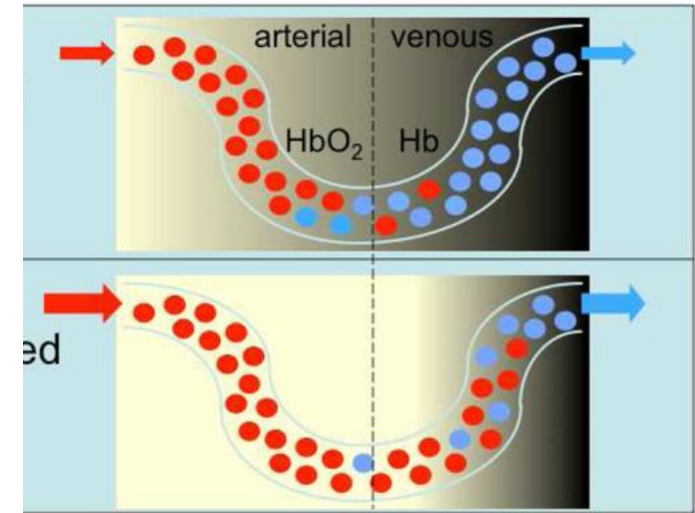
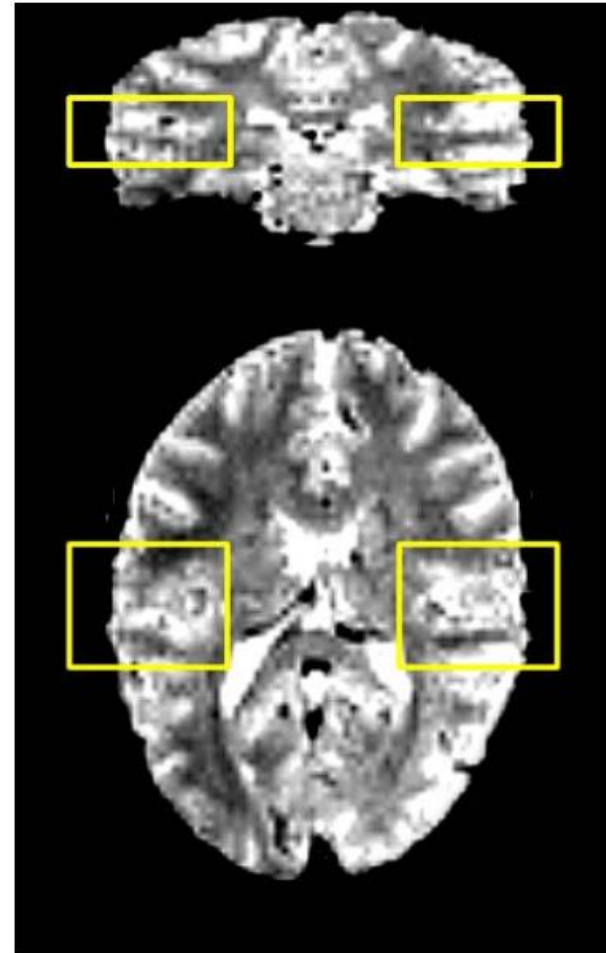


Blood Oxygen Level-Dependent (BOLD) signal

Baseline



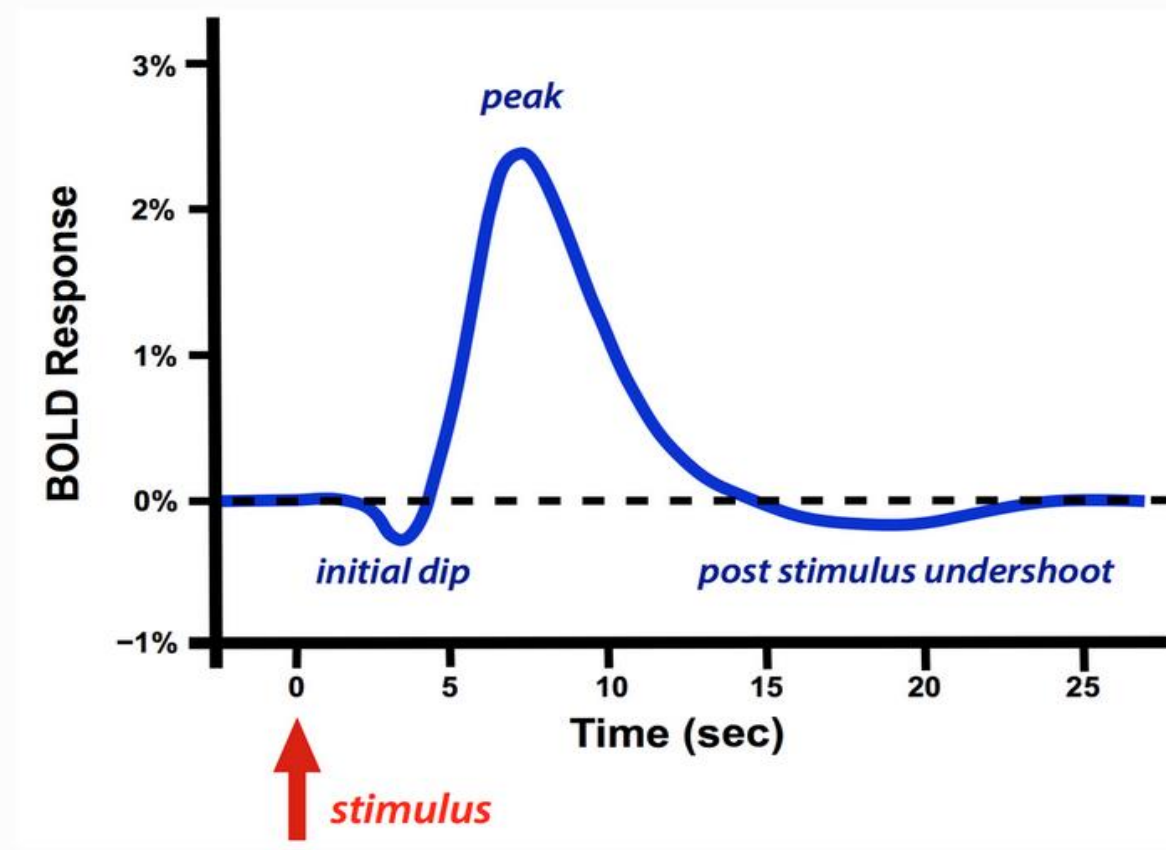
Neural Activity



BOLD response

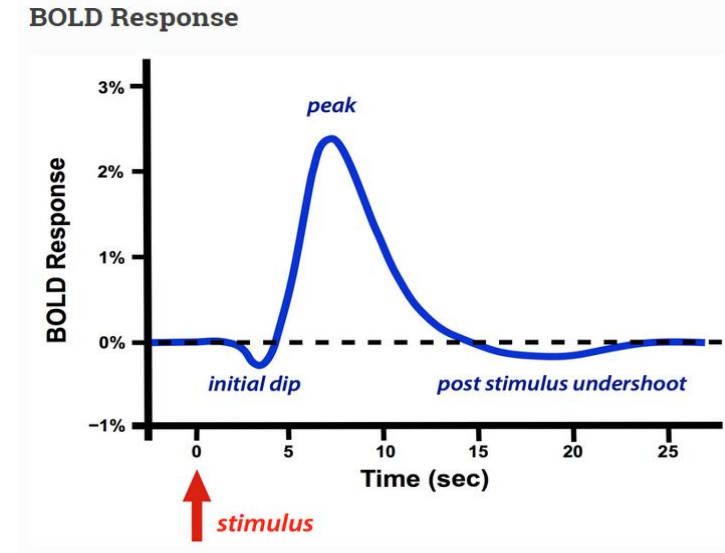
Hemodynamic response function (HRF)

BOLD Response



Hemodynamic response function (HRF)

- Depends on stimulus intensity and duration
- Varies across individuals
- Varies with healthy ageing and development
- Varies with common stimulants such as caffeine
- Varies across the brain, both at a distant and local scale
- The most common solution to HRF variability is to pretend it doesn't exist and use a generic model for all participants



SCIENTIFIC DATA

OPEN

SUBJECT CATEGORIES

- » Electroencephalography
-EEG
- » Brain imaging
- » Functional magnetic
resonance imaging
- » Cognitive neuroscience

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Accepted: 05 January 2015

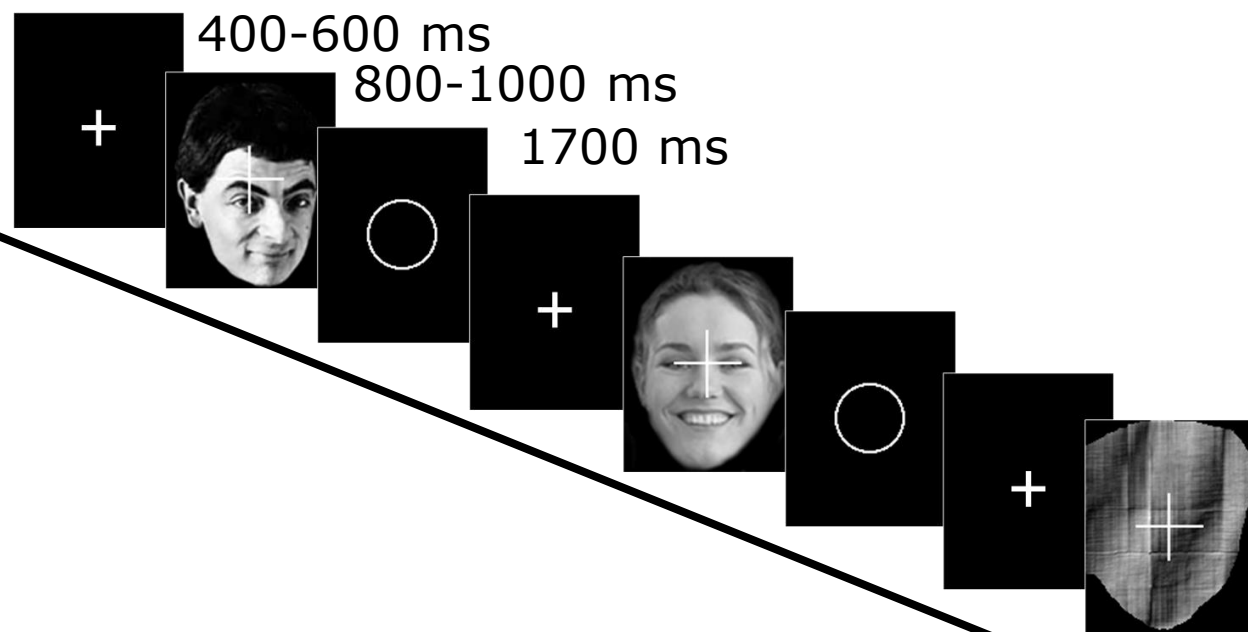
Published: 20 January 2015

A multi-subject, multi-modal human neuroimaging dataset

Daniel G. Wakeman^{1,2} & Richard N. Henson²

We describe data acquired with multiple functional and structural neuroimaging modalities on the same nineteen healthy volunteers. The functional data include Electroencephalography (EEG), Magnetoencephalography (MEG) and functional Magnetic Resonance Imaging (fMRI) data, recorded while the volunteers performed multiple runs of hundreds of trials of a simple perceptual task on pictures of familiar, unfamiliar and scrambled faces during two visits to the laboratory. The structural data include T1-weighted MPAGE, Multi-Echo FLASH and Diffusion-weighted MR sequences. Though only from a small sample of volunteers, these data can be used to develop methods for integrating multiple modalities from multiple runs on multiple participants, with the aim of increasing the spatial and temporal resolution above that of any one modality alone. They can also be used to integrate measures of functional and structural connectivity, and as a benchmark dataset to compare results across the many neuroimaging analysis packages. The data are freely available from <https://openfmri.org/>.

Example Experiment: Face Recognition



N = 16 subjects

Stimuli: 3 types of greyscale face images:

~150 x Familiar

~150 x Unfamiliar

~150 x Scrambled

Task: Judge face symmetry

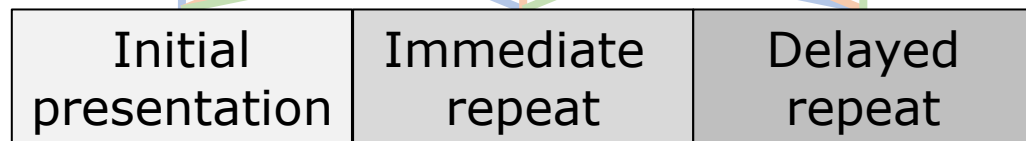
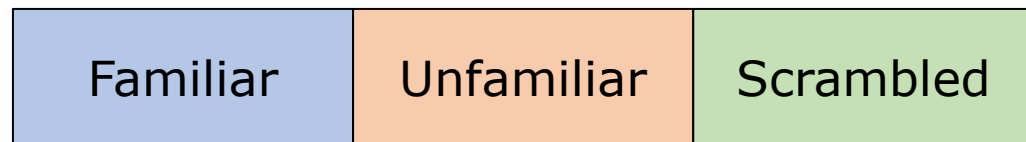
7 min long runs
9 runs
20s Rest after ever 50s

Each image was **presented twice**, with the second presentation occurring either immediately after (**Immediate Repeats**), or after 5–15 intervening stimuli (**Delayed Repeats**), with 50% of each type of repeat.

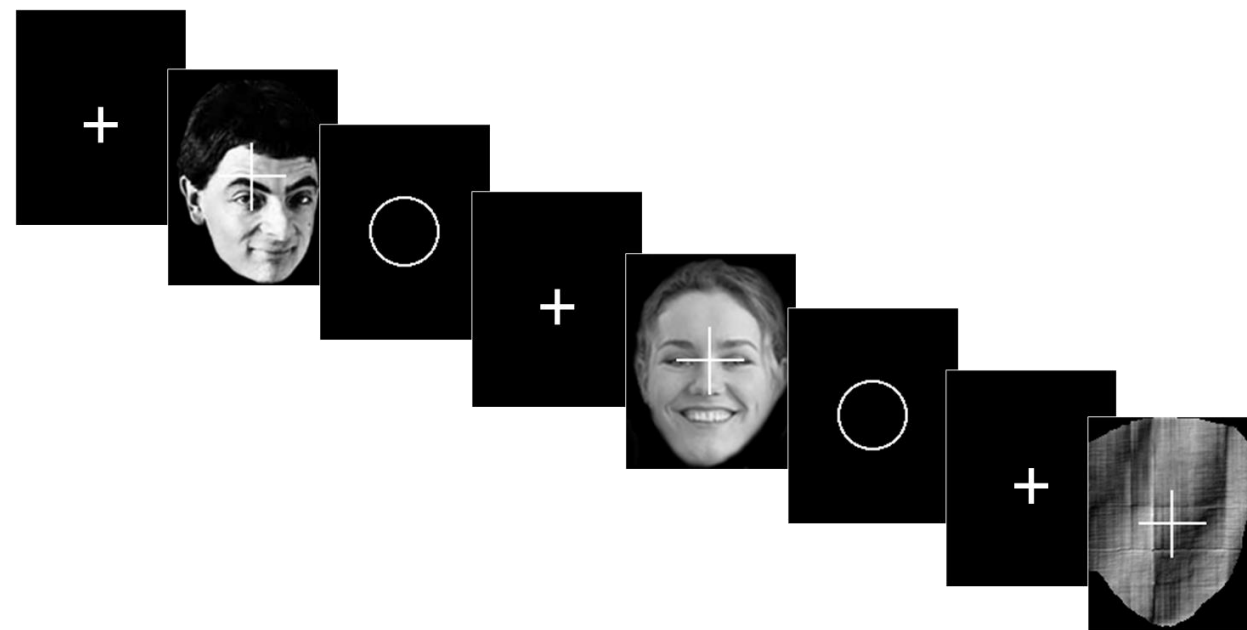
Example Experiment: Face Recognition

- Nine Conditions (3 x 3)

Face



Presentation

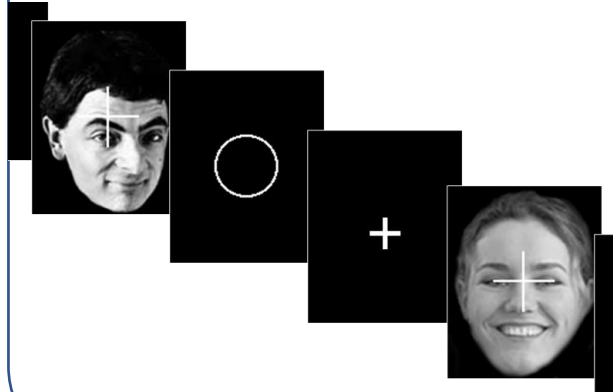


Possible questions to investigate

- Brain areas for Faces
- Brain areas for Face Familiarity
- Response to Initial vs Repeated presentations
- Response to the Repetition of Familiar vs Repetition of Unfamiliar
- ...

An fMRI study

Design an experiment



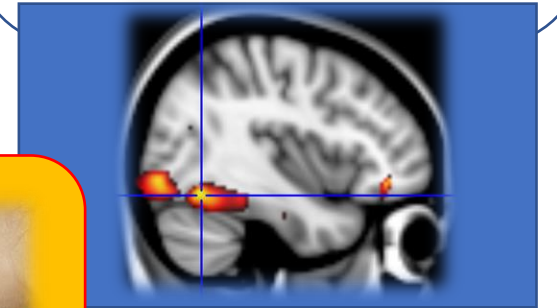
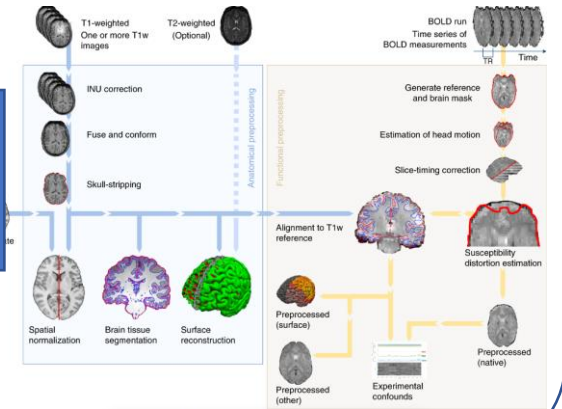
Collect the MRI data



Data

Anatomical image
Functional images
Event details

Pre-process & Analyse



An fMRI study

Design an experiment



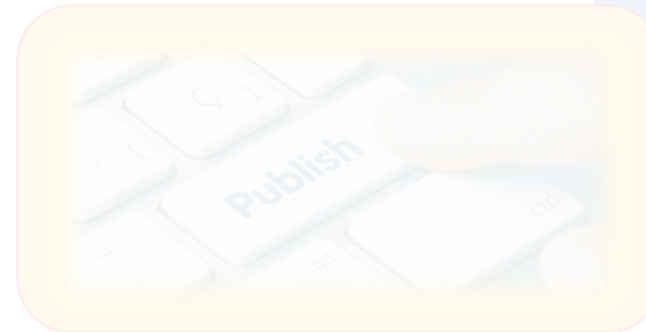
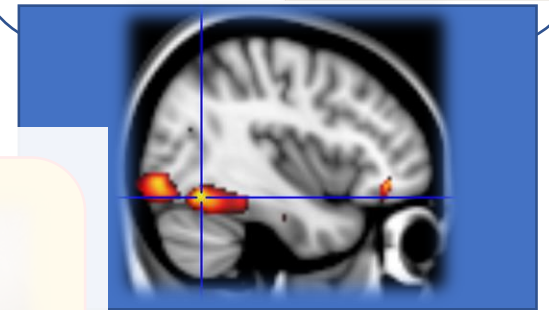
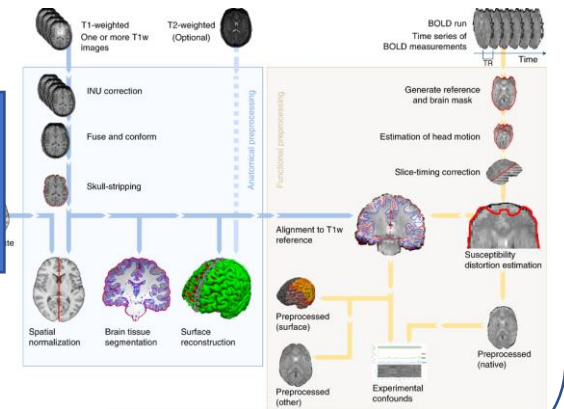
Collect the MRI data



Data

Anatomical image
Functional images
Event details

Pre-process & Analyse



Environment

Data

Organise & Manage

Pre-process

Analyse

Report

Environment

PROGRAMMING LANGUAGES



BASH
&
Shell Scripts

A low-level programming language providing a command line user interface for Unix-like operating systems (e.g., Linux, macOS).

Used to automate repetitive tasks and manage system processes and resources.



A high-level, general-purpose programming language.

License-free – good for reproducible & open code.



MATLAB[®]

A high-level programming language designed for engineers and scientists.

Requires a license. Provides loads of useful resources for Neuroimaging analysis.



Environment

ING LANGUAGES

PACKAGE MANAGER



Conda is an open-source, cross-platform, language-agnostic package manager and environment management system.

With conda, you can use environments that have different versions of Python and packages installed in them.

You can, for example, create your **MRI analysis environment** that includes packages needed for your analysis work.

```

1  name: mri
2  channels:
3    - conda-forge
4    - defaults
5  dependencies:
6    - dcm2niix=1.0.20250506
7    - heudiconv=1.3.3
8    - pip=24.2
9    - pytest=8.3.2
10   - python=3.11.10 # dipy v1.9.0 dependency >=3.9, <3.12
11   - seaborn=0.13.2
12   - traits=6.4.3
13   - wheel=0.44.0
14   - pip:
15     - antspyx==0.5.3
16     - atlasreader==0.3.2
17     - dipy==1.9.0
18     - dcmstack==0.9
19     - fury==0.11.0 # dipy v1.9.0 visualisation didn't work with fury 0.10.
20     - ipykernel==6.29.3
21     - ipython==8.22.1
22     - jupyter==1.0.0
23     - matplotlib==3.8.3
24     - nibabel==5.3.1
25     - nilearn==0.12.0
26     - nipype==1.10.0
27     - nipy==0.6.0
28     - numpy==1.26.4 # dipy v1.9.0 dependency >=1.21.6, <1.27.0
29     - nxviz==0.7.4
30     - pandas==2.2.2
31     - plotly==5.23.0
32     - pybids==0.19.0
33     - python-louvain==0.16
34     - requests==2.31.0
35     - rsatoolbox==0.1.5
36     - scikit-image==0.24.0

```

Environment

PACKAGE MANAGER



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Environment

PROGRAMMING LANGUAGES



BASH
&
Shell Scripts



PACKAGE MANAGER



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With conda, you can use environments that have different versions of Python and packages installed in them.

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<https://conda.io/projects/conda/en/latest/user-guide/tasks/manage-environments.html>

Creating an environment from a .yaml file

```
conda env create -f mri_environment.yaml
```

✓ Environment

Data

Organise & Manage

Pre-process

Analyse

Report

```

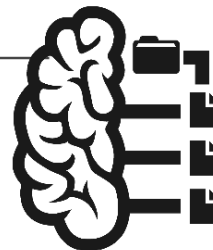
mridata/
├── CBU090962_MR09029
│   └── 20090902_100102
│       ├── Series_001_CBU_Localiser
│       ├── Series_002_CBU_MPRAGE
│       ├── Series_003_CBU_DWEPI_BOLD210
│       ├── Series_004_CBU_DWEPI_BOLD210
│       ├── Series_005_CBU_DWEPI_BOLD210
│       ├── Series_006_CBU_DWEPI_BOLD210
│       ├── Series_007_CBU_DWEPI_BOLD210
│       ├── Series_008_CBU_DWEPI_BOLD210
│       ├── Series_009_CBU_DWEPI_BOLD210
│       ├── Series_010_CBU_DWEPI_BOLD210
│       ├── Series_011_CBU_DWEPI_BOLD210
│       ├── Series_012_CBU_FieldMapping
│       └── Series_013_CBU_FieldMapping

```

```

sub-15
├── ses-mri
│   ├── anat
│   │   ├── sub-15_ses-mri_T1w.json
│   │   └── sub-15_ses-mri_T1w.nii.gz
│   ├── fmap
│   │   ├── sub-15_ses-mri_acq-func_magnitude1.json
│   │   ├── sub-15_ses-mri_acq-func_magnitude1.nii.gz
│   │   ├── sub-15_ses-mri_acq-func_magnitude2.json
│   │   ├── sub-15_ses-mri_acq-func_magnitude2.nii.gz
│   │   ├── sub-15_ses-mri_acq-func_phasediff.json
│   │   └── sub-15_ses-mri_acq-func_phasediff.nii.gz
│   └── func
│       ├── sub-15_ses-mri_task-facerecognition_run-01_bold.json
│       ├── sub-15_ses-mri_task-facerecognition_run-01_bold.nii.gz
│       ├── sub-15_ses-mri_task-facerecognition_run-01_events.tsv
│       ├── sub-15_ses-mri_task-facerecognition_run-02_bold.json
│       ├── sub-15_ses-mri_task-facerecognition_run-02_bold.nii.gz
│       ├── sub-15_ses-mri_task-facerecognition_run-02_events.tsv
│       ├── sub-15_ses-mri_task-facerecognition_run-03_bold.json
│       ├── sub-15_ses-mri_task-facerecognition_run-03_bold.nii.gz
│       ├── sub-15_ses-mri_task-facerecognition_run-03_events.tsv
│       ├── sub-15_ses-mri_task-facerecognition_run-04_bold.json
│       ├── sub-15_ses-mri_task-facerecognition_run-04_bold.nii.gz
│       ├── sub-15_ses-mri_task-facerecognition_run-04_events.tsv
│       ├── sub-15_ses-mri_task-facerecognition_run-05_bold.json
│       ├── sub-15_ses-mri_task-facerecognition_run-05_bold.nii.gz
│       ├── sub-15_ses-mri_task-facerecognition_run-05_events.tsv
│       ├── sub-15_ses-mri_task-facerecognition_run-06_bold.json
│       ├── sub-15_ses-mri_task-facerecognition_run-06_bold.nii.gz
│       ├── sub-15_ses-mri_task-facerecognition_run-06_events.tsv
│       ├── sub-15_ses-mri_task-facerecognition_run-07_bold.json
│       ├── sub-15_ses-mri_task-facerecognition_run-07_bold.nii.gz
│       ├── sub-15_ses-mri_task-facerecognition_run-07_events.tsv
│       ├── sub-15_ses-mri_task-facerecognition_run-08_bold.json
│       ├── sub-15_ses-mri_task-facerecognition_run-08_bold.nii.gz
│       ├── sub-15_ses-mri_task-facerecognition_run-08_events.tsv
│       ├── sub-15_ses-mri_task-facerecognition_run-09_bold.json
│       ├── sub-15_ses-mri_task-facerecognition_run-09_bold.nii.gz
│       ├── sub-15_ses-mri_task-facerecognition_run-09_events.tsv
│       └── sub-15_ses-mri_scans.tsv

```



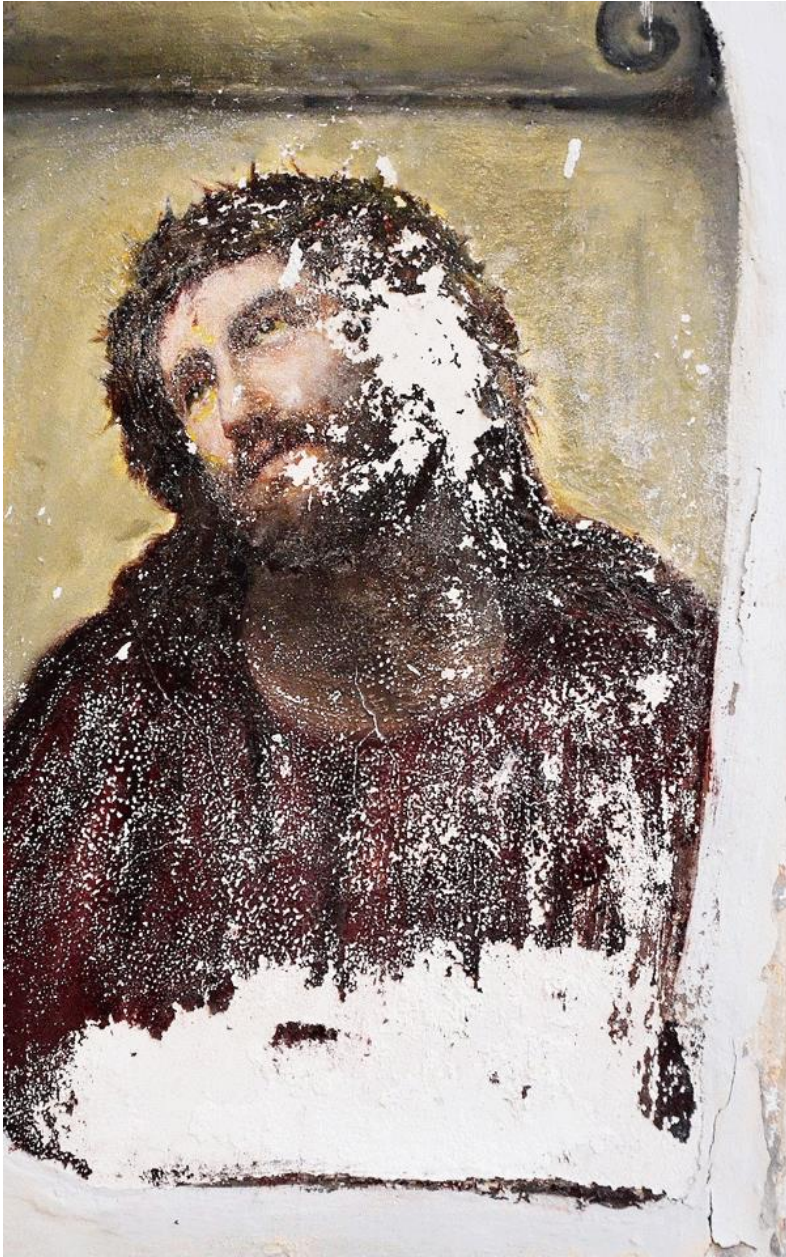
✓ Environment



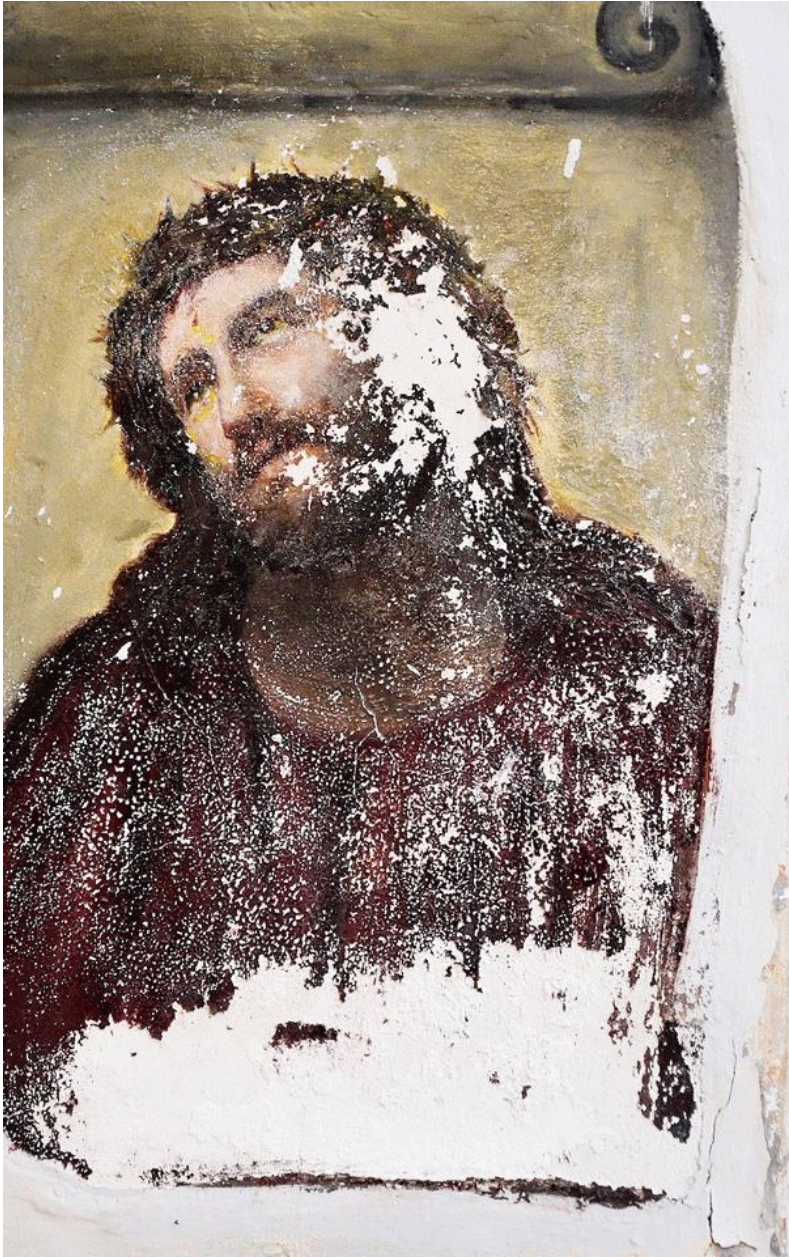
Data
Organise & Manage

Pre-process

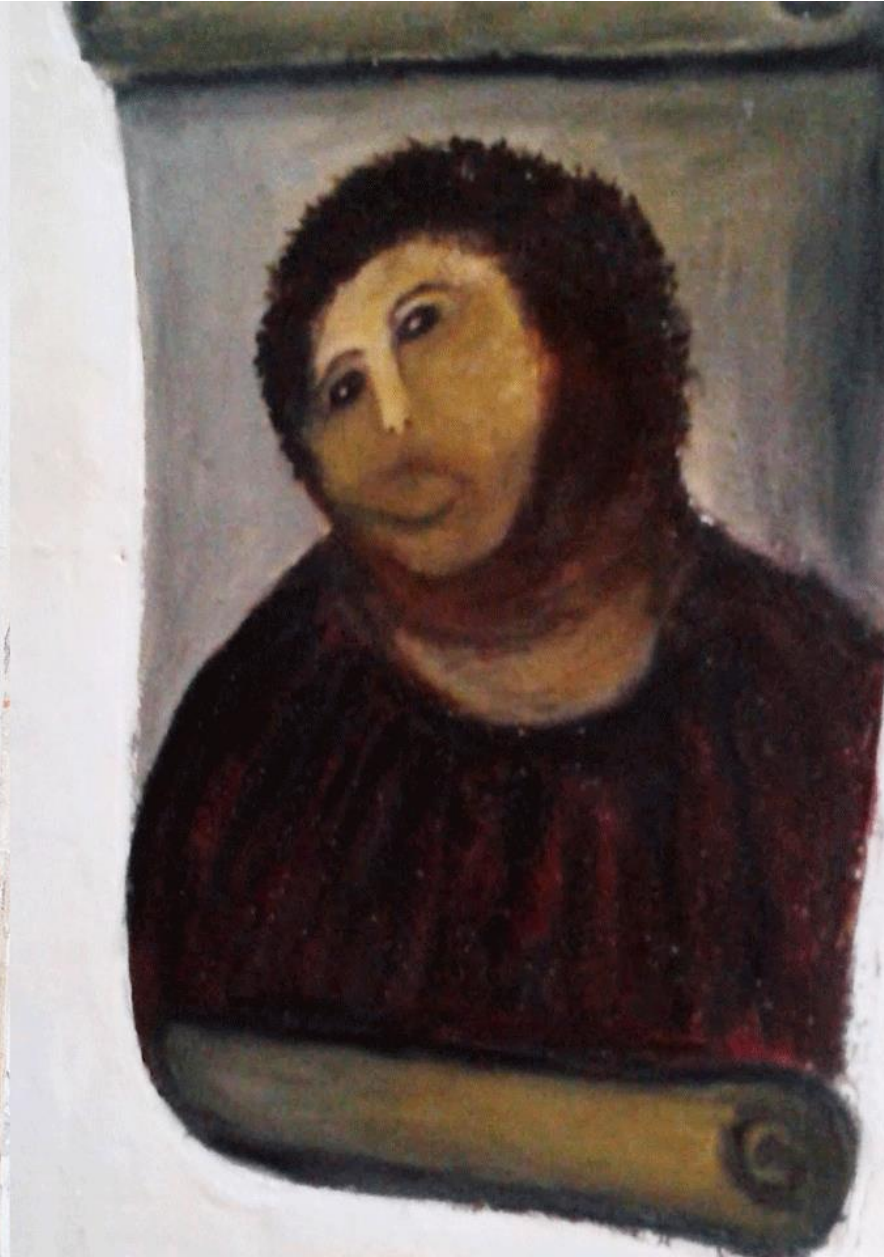
raw



raw

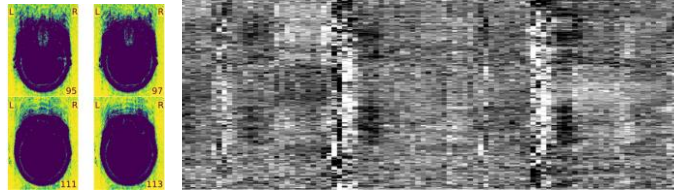


pre-processed

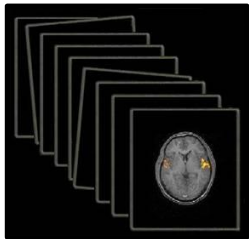


Typical fMRI pre-processing pipeline

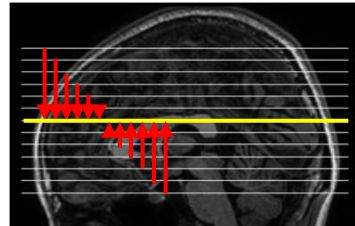
Data quality
assessment



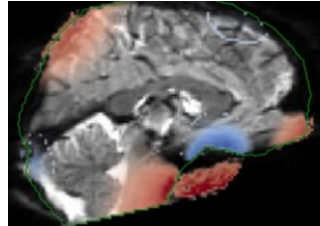
Motion-
correction



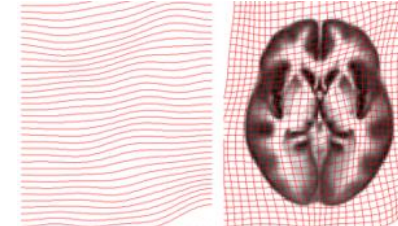
Slice-time
correction



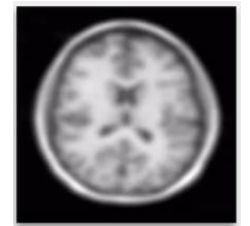
Distortion
correction



Normalisation to
MNI space

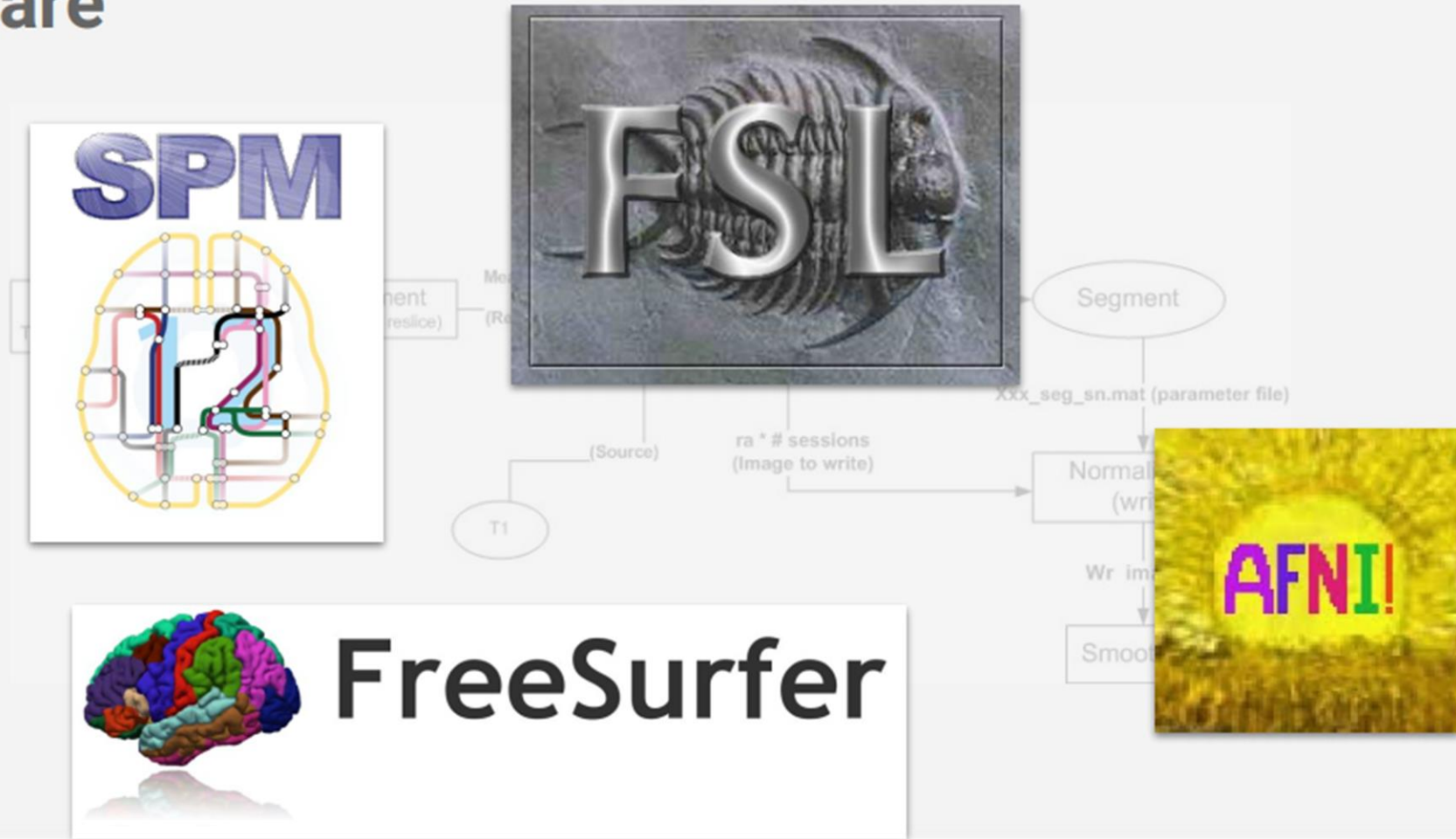


Spatial
smoothing

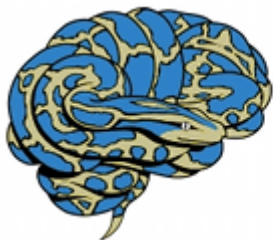


Pre-processing tools

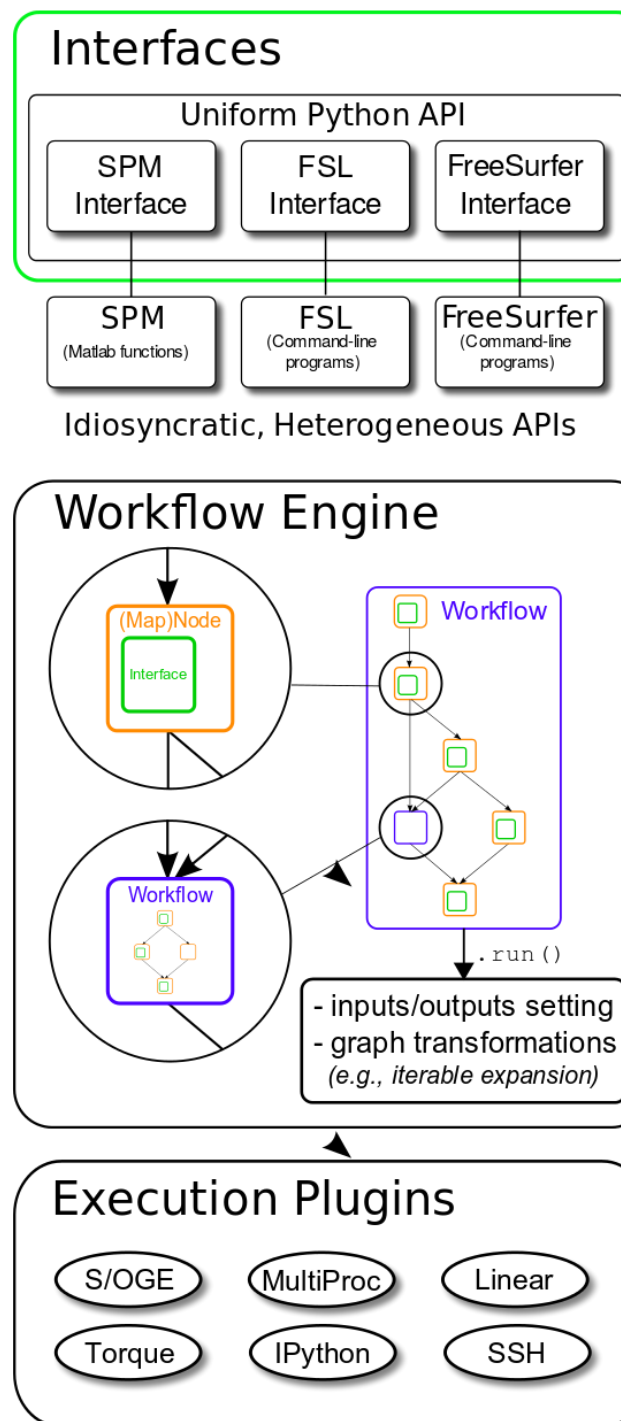
Software



Pre-processing tools



Nipype: Neuroimaging in Python Pipelines and Interfaces



Pre-processing tools

- **fMRIPrep** <https://fmriprep.org/en/stable/>
 - Fully automated fMRI data pre-processing tool
 - The workflow is based on [Nipype](#) and encompasses a large set of tools from well-known neuroimaging packages, including [FSL](#), [ANTs](#), [FreeSurfer](#), [AFNI](#), and [Nilearn](#). This pipeline is designed to provide the best software implementation for each state of pre-processing.
 - **Robustness** - The pipeline adapts the pre-processing steps depending on the input dataset and should provide results as good as possible independently of scanner make, scanning parameters or presence of additional correction scans (such as fieldmaps).
 - **Ease of use** - Thanks to dependence on the BIDS standard, manual parameter input is reduced to a minimum, allowing the pipeline to run in an automatic fashion.
 - **“Glass box” philosophy** - Automation should not mean that one should not visually inspect the results or understand the methods. Thus, fMRIPrep provides visual reports for each subject, detailing the accuracy of the most important processing steps.



Environment



Data
Organise & Manage



Pre-process

Analyse

Finding a **Face** area in the brain

Which brain regions are engaged when people look at faces



- With fMRI the meaningful questions are questions that **compare two conditions**
 - We need some sort of control condition → **Scrambled** condition
- Which brain regions respond more to looking at **face** images than **scrambled** images
 - The control question hopefully helps to **wash out all the regions we are not interested in**. Because regions that we are NOT interested in should activate both conditions to the same extent (e.g. visual areas)

GLM: $Y = X\beta + \varepsilon$

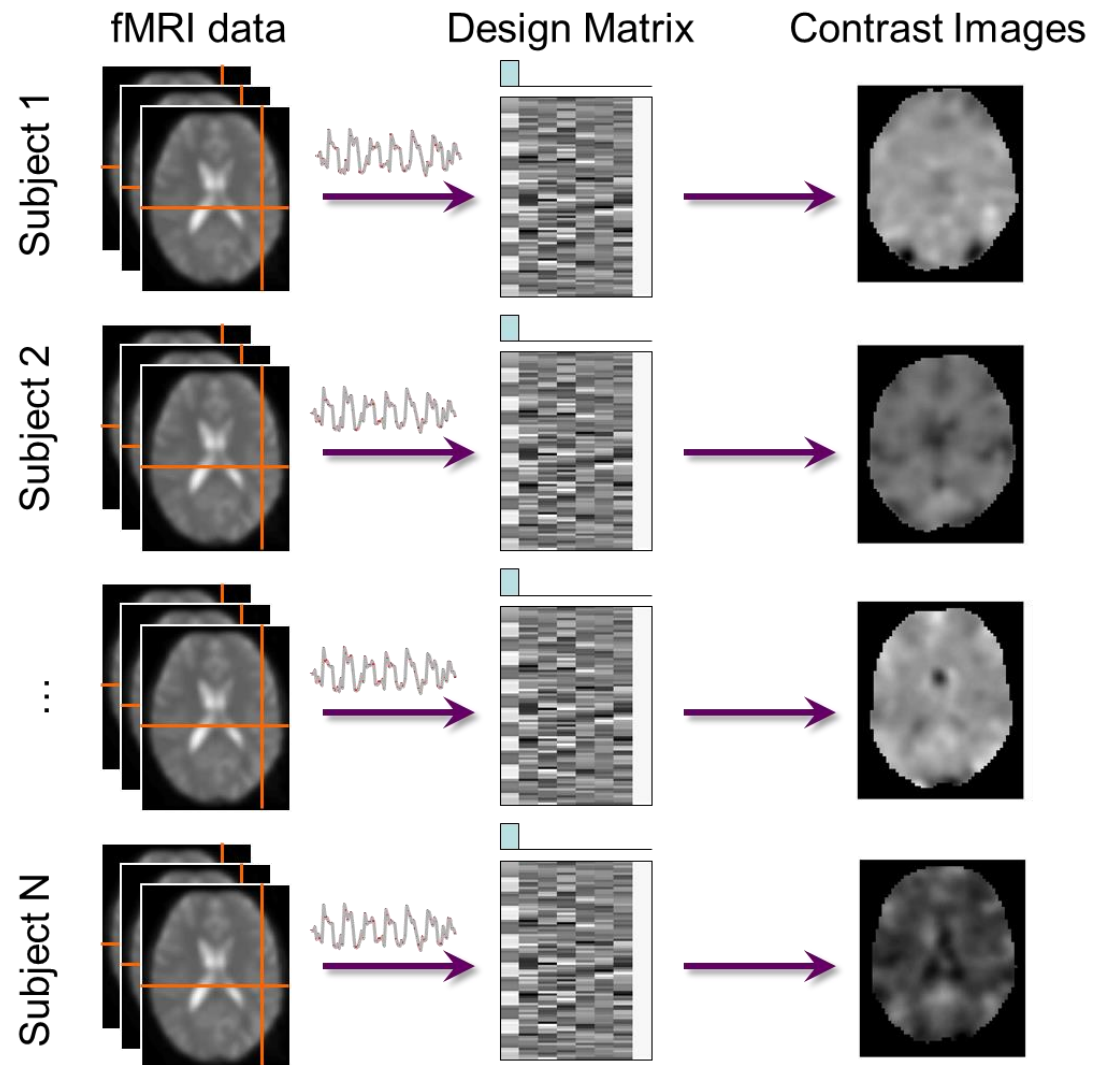
$$\text{BOLD signal} = \underbrace{X * b}_{\substack{\text{explained variation} \\ \text{task-related activity changes}}} + \underbrace{\text{errors}}_{\substack{\text{unexplained variation} \\ \text{noise (other changes)}}}$$

- **What we know?**
 - **BOLD signal**: we collect this from the brain (functional data)
 - **X**: the design matrix (each column is a predictor that we build ourselves)
- **What we want to find?**
 - **b**: vector of beta-weights (one weight for predictor in X) that give the best approximation of the BOLD signal
- **How we find it?**
 - By **minimising the sum of squared errors**. In practice, the **GLM** has a formula, which guarantees to find these beta-weights

1. **Extract the signal time-series** from a given voxel
2. **Run GLM** (the signal and the design matrix are the inputs) to **find beta-weights** that best approximate the true signal
3. Define your **contrast** and test it
4. Repeat for **all voxels**
 - Produces an image file with contrast values for each voxel: **contrast-maps**

First-level analysis

- Run the GLM for each subject



Group level (2nd level) analysis is across subjects

- Which voxels are showing significant activation differences between our conditions consistently **within a group**
- Importantly, all subject brains need to be in a common space, e.g. MNI, to perform voxel-wise group analyses



Environment



Data Organise & Manage



Pre-process



Analyse

Report

Sharing & Reporting



- Share your **code** and notebooks on GitHub



- Make it **citable** with Zendono
 - <https://docs.github.com/en/repositories/archiving-a-github-repository/referencing-and-citing-content>



- If you have consent from participants, share the **BIDS data** on OpenNeuro.

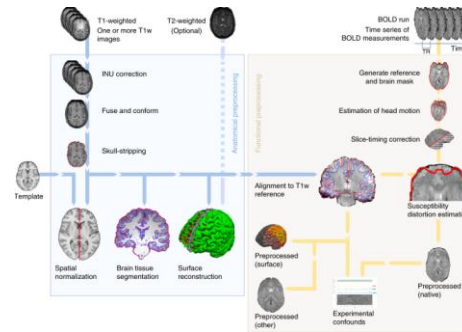


- Add your **contrast maps** to NeuroVault

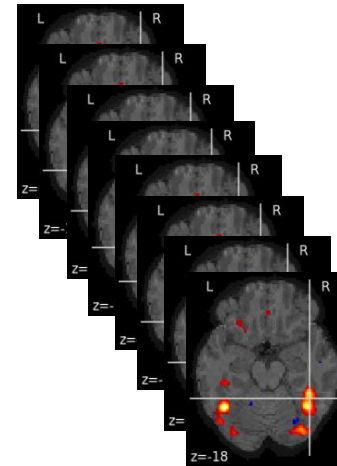
Data Organisation

MRI Data Manipulation

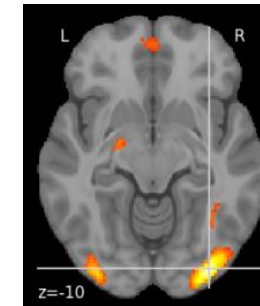
Quality Control & Pre-processing



Subject-Level Analysis



Group-Level Analysis



ROI Analysis

